RNS INSTITUTE OF TECHNOLOGY

Autonomous Institution Affiliated to VTU
RR Nagar, Bengaluru-560098

I / II Semester B.E. Degree Examination – June / July 2026
Building Sciences and Mechanics

Time: 3 hrs **Max. Marks: 100**

Note:

1. ***Answer any 5 full questions, selecting at least one question from each module.***
2. ***Missing data may be suitably assumed.***
3. ***M: Marks, L: Bloom's level, C: Course Outcomes***

Q.No.		Module-1	M	L	C
Q1	a	Explain briefly the scopes of a) Transportation Engineering b) Geotechnical Engineering	8	L2	CO1
	b	Enumerate the application of the following building materials. a) Bricks b) Cement	8	L2	CO1
	c	Define Plinth Beam and explain its uses.	4	L2	CO1
		OR			
Q2	a	Explain in detail Masonry wall and its functions.	8	L2	CO1
	b	Define foundation and explain its classification with sketches.	8	L2	CO1
	c	Define staircase and list its types.	4	L2	CO1
		Module-2			
Q3	a	Discuss the properties of bamboo that make it a suitable material for sustainable construction.	10	L2	CO2
	b	Explain in detail smart city concept.	10	L2	CO2
		OR			
Q4	a	Explain the IGBC rating system for new buildings – purpose and main features.	10	L2	CO2
	b	What is GRIHA (Green Rating for Integrated Habitat Assessment)? Discuss its role in India's sustainable development.	10	L2	CO2
		Module-3			
Q5	a	Define the resolution and composition of forces.	4	L2	CO2
	b	State and prove Varignon's theorem of moments.	8	L2	CO2

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	c	Determine the magnitude and direction of the resultant force for the force system shown in Fig. Q5(c).	8	L3	CO3

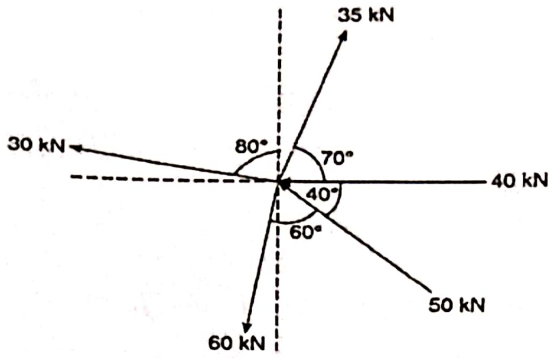


Fig. Q5(c)

OR

Q6	a	Define a couple and mention its characteristics.	5	L2	CO2
	b	State and explain the principle of transmissibility of forces	5	L2	CO2
	c	Determine the magnitude, direction and position of the resultant force with reference to the point A for the non-coplanar force system shown in Fig. Q6(c).	10	L3	CO3

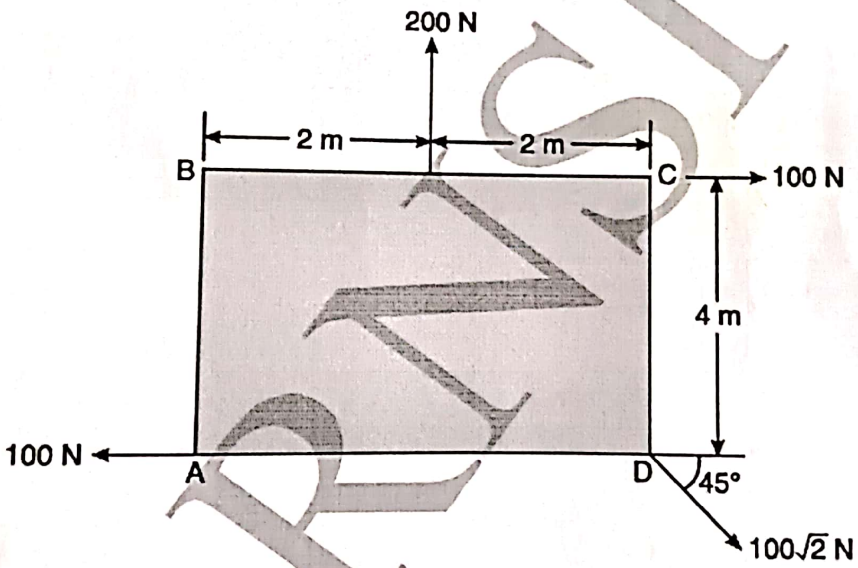


Fig. Q6(c)

Module-4

Q7	a	State the conditions of static equilibrium for coplanar concurrent and coplanar non-concurrent force systems.	6	L2	CO2
	b	State and prove Lami's theorem.	4	L2	CO2
	c	Two identical rollers, each weighing 400 N, are placed in a trench as shown in Fig. Q7(c). Assuming that all contact surfaces are smooth, determine the reactions at contact points A, B, C, and D.	10	L3	CO3

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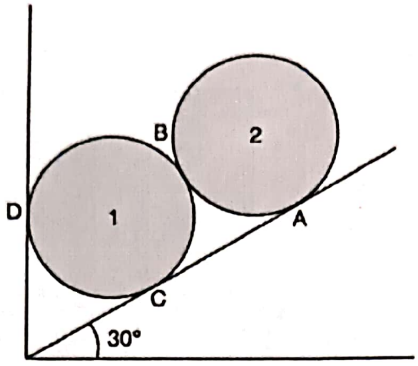
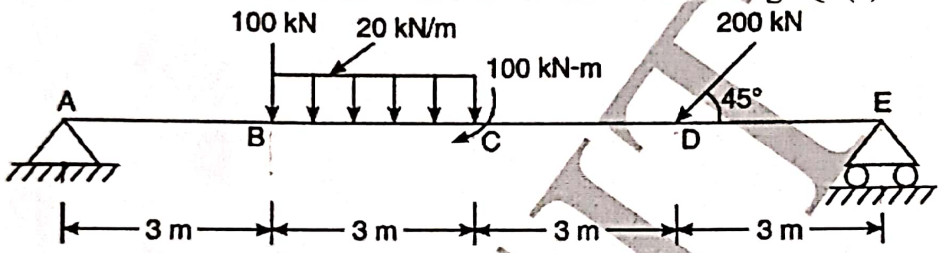
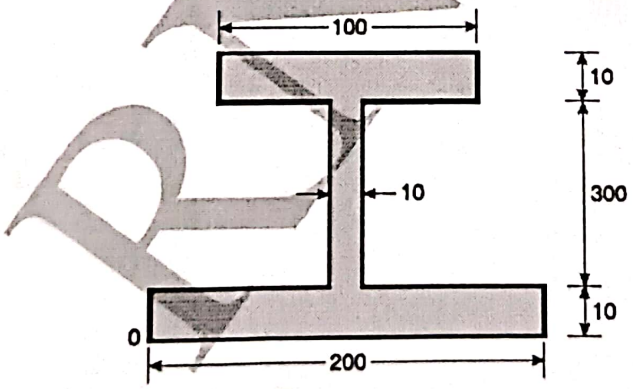


Fig. Q7(c)

OR

Q8	a	Differentiate between statically determinate and indeterminate beams.	6	L2	CO2
	b	Explain different types of beams with neat sketches	6	L2	CO2
	c	Determine the reactions at A and E for the beam shown in Fig. Q8(c).  Fig. Q8(c)	8	L3	CO3

Module-5

Q9	a	Derive the expression to locate the centroid of a rectangle.	10	L2	CO
	b	Determine the position of centroid with respect to 'O' shown in Fig.Q9(b) All dimensions are in mm.  Fig.Q9(b)	10	L3	CO

OR

Q10	a	Derive an expression to locate the centroid of a Triangle.	10	L2	CO
	b	Determine the position of the centroid of the shaded area shown in Fig.Q10(b).	10	L3	CO

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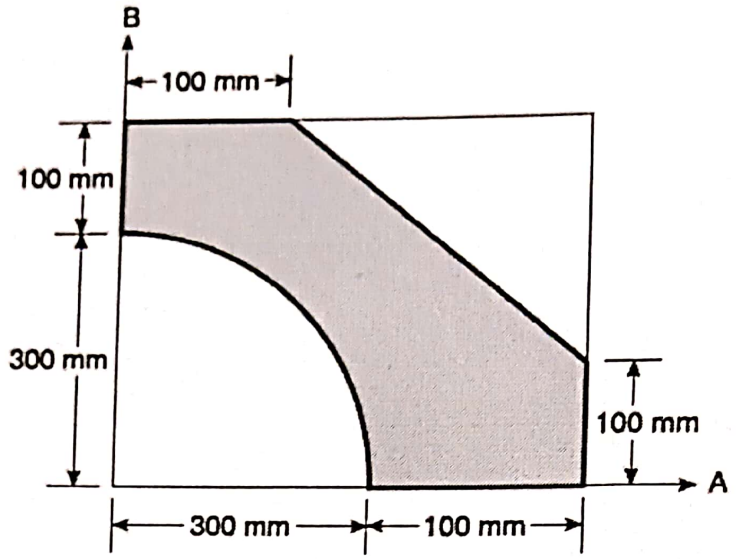


Fig.Q10(b)

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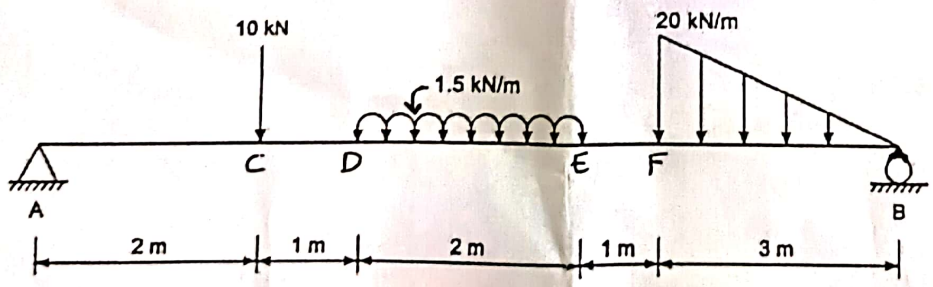
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CIE – Test II

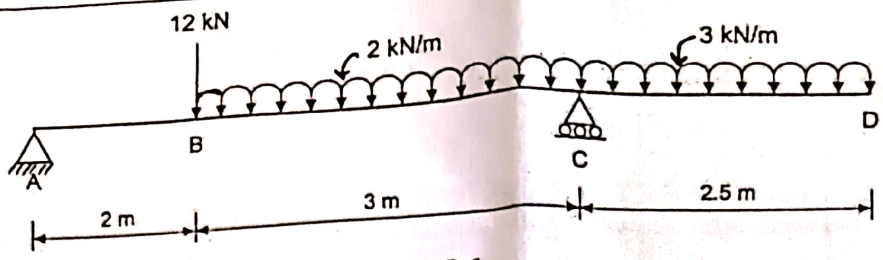
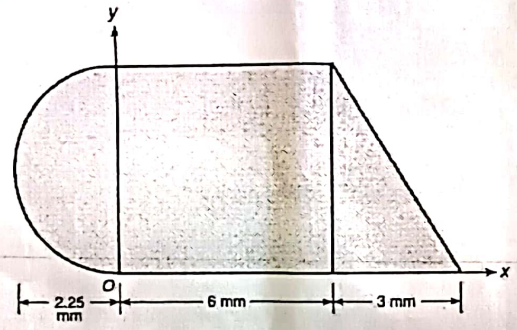
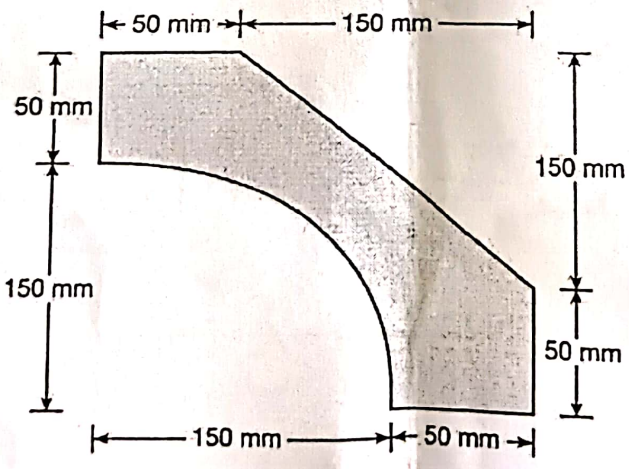
Sem: II
Subject: Building Sciences and Mechanics
Subject Code: BENSC204A

Date: 30-05-2026
Time: 9.30 am – 11.00 am
Max. Marks: 50

Instruction: Answer any 5 full questions, selecting ONE question from each part.

Q.No.		Question	Marks	RBT*	COs
1	a	Enumerate various types of Autoclaved Aerated Concrete (AAC) blocks with their uses.	05	L2	CO2
	b	What is smart city and explain the Benefits of Smart Cities.	05	L2	CO2
		OR			
2	a	Describe the Sustainability Criteria for Emerging Materials.	05	L2	CO2
	b	Write the Advantages of Bamboo as a Building Material.	05	L2	CO2
3	a	What is green building and write the Advantages of Green Buildings for Sustainable Living.	05	L2	CO2
	b	Briefly explain GRIHA (Green Rating for Integrated Habitat Assessment).	05	L2	CO2
		OR			
4	a	Explain the concept and purpose of Green Building Rating Systems.	05	L2	CO2
	b	Briefly write the Challenges in implementation of Green Building.	05	L2	CO2
5	a	Determine the reactions at A and B for the beam shown in Fig. Q5.  **Fig.Q5.**	07	L3	CO3
	b	List with neat sketch of different types of supports.	03	L2	CO3
		OR			
6	a	Determine the reaction at support A & C for the system shown in Fig. Q6.	07	L3	CO3

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		 Fig.Q6.			
	b	List with neat sketch various types of loads on beams.	03	L2	CO3
7		Derive an expression for locating the centroid of a rectangle using the method of integration.	10	L2	CO4
		OR			
8		Derive an expression for locating the centroid of a triangle using the method of integration.	10	L2	CO4
9		Determine the centroid of the given Fig. Q9.  Fig.Q9.	10	L3	CO4
		OR			
10		Determine the centroid of the area shown in Fig. Q10, for the shaded portion.  Fig.Q10.	10	L3	CO4

*** Revised Bloom's Taxonomy: L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create**

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RNS INSTITUTE OF TECHNOLOGY

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I Semester B.E. Degree Examination – Jan / Feb 2026
Introduction to Mechanical Engineering

Time: 3 hrs **Max Marks:100**

Note:

1. ***Answer any 5 full questions, selecting at least one question from each module.***
2. ***Missing data may be suitably assumed.***
3. ***M: Marks, L: Bloom's level, C: Course Outcomes***

Q.No.		Module-1	M	L	C
Q1	a	Differentiate between renewable and non-renewable energy resources.	10	L2	CO1
	b	Explain with the neat sketch , the working of hydel power plant	10	L2	CO1
		OR			
Q2	a	Explain with the neat sketch ,the working of Pelton Wheel/Turbine	10	L2	CO1
	b	Explain the parts of refrigeration system	10	L2	CO1
		Module-2			
Q3	a	Explain with neat sketch, the following operations performed on Lathe Machine i)Turning ii) Facing iii)Knurling	10	L2	CO2
	b	Explain with neat sketch, the following operations performed on Drilling Machine i)Drilling ii)Boring iii) Reaming	10	L2	CO2
		OR			
Q4	a	Explain the block diagram of CNC Machine	10	L2	CO2
	b	Explain with neat sketch, FDM in 3D Printing	10	L2	CO2
		Module-3			
Q5	a	Explain with the neat sketch, the working principle of 4-stroke Petrol Engine	10	L2	CO3
	b	Explain with the neat sketch, the working principle of 4-stroke Diesel Engine	10	L2	CO3
		OR			
Q6	a	Explain the basic components of Electric Vehicle	10	L2	CO3
	b	Explain the advantages and disadvantages of Hybrid Electric Vehicle	10	L2	CO3
		Module-4			
Q7	a	List advantages and applications of composite materials	10	L2	CO4
	b	Explain the type & need of reinforcement materials in composites.	10	L2	CO4
		OR			
Q8	a	With a neat sketch explain Arc welding principle	10	L2	CO4
	b	Explain different types of flames used in gas welding.	10	L2	CO4

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		Module-5			
Q9	a	Explain robot configuration: polar and cylindrical type.	10	L2	CO5
	b	Explain working principle of potentiometer and capacitive sensor.	10	L2	CO5
		OR			
Q10	a	Explain the salient features of different types of Automation.	10	L2	CO5
	b	Define IIoT and explain its characteristics.	10	L2	CO5

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CIE – Test 2

Sem: I
Subject: Introduction to Mechanical Engineering (P Cycle)
Subject Code: BENSC104D

Date: 31/12/2025
Time: 9:30-11:00
Max. Marks: 50

Instruction: Answer any 5 full questions, selecting ONE question from each part.

Q.No.		Question	Marks	RBT	COs
1	a	Explain with sketch the components of Electric vehicle.	10	L2	CO3
		OR			
2	a	Explain with neat sketch simple and compound gear train.	10	L2	CO3
3	a	Explain types and its applications of Ferrous and Non-Ferrous metals.	10	L2	CO4
		OR			
4	a	Enlist the advantages, disadvantages and applications of Composite materials	10	L2	CO
5	a	Explain with neat sketch the working principle of Arc welding.	10	L2	CO
		OR			
6	a	Define gas welding. Explain the types of flames used in Gas welding.	10	L2	CO
7	a	Define Mechatronics. Explain the elements of Mechatronic system.	10	L2	CO
		OR			
8	a	Explain with sketch four configuration of Robot.	10	L2	CO
9	a	Explain the types of Automation.	10	L2	CO
		OR			
10	a	Explain the characteristics of IIoT and Industry 4.0	10	L2	CO

*** Revised Bloom's Taxonomy: L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create**

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Quiz

1. Regenerative braking in electric vehicles helps to: A. Increase brake wear B. Reduce motor speed only C. Convert kinetic energy into electrical energy D. Convert electrical energy into heat	**2. One major environmental benefit of electric vehicles is:** A. Increased noise pollution B. Zero tailpipe emissions C. High exhaust emissions D. Fuel leakage
3. Which of the following is a ferrous material? A. Brass B. Aluminium C. Cast iron D. Copper	**4. A composite material is made by combining:** A. Two similar materials B. Two or more dissimilar materials C. Only metallic materials D. Only polymeric materials
5. Welding is a process of joining materials by A. Mechanical fastening B. Adhesive bonding C. Fusion with or without pressure D. Riveting	**6. Brazing differs from soldering mainly in terms of** A. Joint strength B. Filler material C. Operating temperature D. Type of base metal
7. Mechatronics is best defined as the integration of: A. Mechanical systems only B. Electrical systems only C. Mechanical, electrical, electronics, and control systems D. Computer systems only	**8. A robot is defined as:** A. A fixed mechanical device B. A manually operated machine C. A reprogrammable, multifunctional manipulator D. A simple automatic machine
9. Which type of automation is suitable for mass production? A. Flexible automation B. Fixed automation C. Programmable automation D. Integrated automation	**10. The main advantage of automation is:** A. Increased human error B. Reduced system reliability C. Consistent product quality D. Increased labour requirement

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